## Stack Overview

| ***Function*** | ***Name*** | ***Dose*** | ***Frequency*** | ***Pattern*** |
|---|---|---|---|---|
| Metabolic | **Retatrutide** | SC 2.4 mg | Sun, Wed | Continuous |
| | **Metformin** | PO 500 mg | Daily | Continuous |
| GH-axis | **Ipamorelin + CJC-1295 no-DAC** | SC 200 mcg + 200 mcg | Sun–Thu PM (fasted) | 12 weeks on / 6 weeks off |
| | **Tesamorelin + Ipamorelin** | SC 2 mg + 200 mcg | M-F AM (fasting) | |
| Tissue repair | **GHK-Cu** | SC 1-2 mg | Daily | Alternating vials |
| | **GLOW** | SC 1.2-2.4 mg | Daily | |
| | **ARA-290** | SC 4, 2, 2 | Tu, Th, Sa | Continuous |
| | **Zinc Picolinate** | PO 30 mg elemental zinc | 3× weekly | Continuous |
| Anti-inflammatory | **KPV** | SC 750 mcg | Daily | Continuous |
| | **Omega-3** | PO DHA 900 mg; DPA 140 mg | Daily, with food | Continuous |
| Immune modulation | **Thymosin-α1** | SC 0.8 mg | M, F | Continuous |
| | **Thymulin** | SC 0.5 mg | Su, Tu, Th | Continuous |
| | **Spermidine 3HCl** | PO 10 mg | Daily (evenings) | Continuous |
| | **Tadalafil** | PO 5 mg | Daily | Continuous |
| | **Multivitamin** | PO 1 capsule | Daily | Continuous |
| | **Minoxidil** | PO 1.25 mg | Daily | Continuous |
| Prostate support | **Beta-sitosterol** | PO 250 mg | Daily, with food | Continuous |
| Neurotrophic | **Semax** | IN 600–900 mcg | Daily | Occasionally |
| Sleep support | **Epitalon + DSIP** | IN 200 + 200 mcg | Daily, 1 hour before bedtime | Continuous |
| | **L-Glycine** | PO 3000 mg | | Continuous |
| Cellular Energy and Brain Support | **NMN** | PO 1000 mg | Daily | Continuous |
| | **Acetyl-L-Carnitine (ALCAR)** | PO 1000 mg | Daily | Continuous |
| | **CoQ10** | PO 200 mg | Daily, with food | Continuous |
| | **Creatine HCl** | PO 1500 mg | Daily | Continuous |
| | **PQQ** | PO 20 mg | Daily | Continuous |
| | **Apigenin** | PO 200 mg | Daily, with food | Continuous |
| | **Beta-alanine** | PO 3000 mg | Daily | Continuous |

These peptides are used infrequently.

| ***Function*** | ***Name*** | ***Dose*** | ***Frequency*** | ***Pattern*** |
|---|---|---|---|---|
| Longevity | **FOXO4-DRI** | SC 25 mg | Q48H ×3 | Every 2 to 3 years |
| | **Epitalon + Crystagen + Pinealon** | SC 5 mg; 0.5 mg; 0.5 mg | Daily for 10 days | Annually |
| Prostate support | **Prostamax (KEDP)** | SC 0.5 mg | Daily for 10 days | Annually |

# Before You Stack

Peptides and supplements are multipliers of your fundamentals: good sleep, good nutrition, and exercise. If your sleep is poor, your nutrition is fast food, your stress is chaotic, or you're not lifting, nothing in a jar, bottle, or vial is going to help much. Get your fundamentals right, then consider your goals and how supplements, peptides, etc. might help.

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

*Evidence classifications used in this document:*

- *Claims supported by good human evidence are simply stated without qualification.*
- *Claims supported by good animal evidence, when not contradicted by human findings, are described as likely.*
- *Claims supported by in vitro or mechanistic evidence, when not contradicted by higher-level findings, are described as plausible.*
- *Claims lacking these forms of support are omitted or, if there is strong community support, identified as speculative.*

*A lack of human evidence may simply reflect that appropriate trials and experiments have not been performed because of cost, limited commercial incentives, weak intellectual-property protection, or the absence of a disease-specific regulatory endpoint. It should not automatically be interpreted as evidence against the biological mechanism or efficacy.*

This document was updated 31-Aug-26. The most recent version of this document is available at https://pdfs.researcher6076.com/pdfs/my-stack.pdf

## Lifestyle Stack

**Nutrition:** I am 100% vegan, focused on whole foods, and limiting (but not fully eliminating) processed food.

**Intermittent Fasting:** I fast five to seven days a week from 7 PM until 11 AM.

**Sleep:** I am focusing on sleep improvements. My bedroom is dark; the bed is temperature controlled for cool sleep. My wake time is the same each day. My average sleep duration is 8 hours at least 5 days/week, but my sleep still needs improvement.

**Exercise:** My resistance training schedule is two to three times each week, consistently. The training is a complete upper and lower body routine with each movement focusing on maximum intensity with 8–20 reps. When 20 reps are consistently achieved, I increase the resistance.

## Stack Goals

My overarching goal is **health support**. The individual goals supporting this are **metabolic stability**, **body composition**, **recovery**, **immune support**, **ageing support and healthspan**. As this stack pursues these individual goals, it **never intentionally wavers from making the overall healthiest choice**.

## Functional Benefits

- **Improved glycemic control, appetite regulation, and body-composition** support through GLP-1/GIP signaling, glucagon-receptor activity, increased energy expenditure, and GH-mediated lipolysis.
- **Enhanced recovery, sleep quality, tissue maintenance,** and **anabolic signaling** through pulsatile GH stimulation.
- **Support for tissue repair, collagen production,** and **microvascular perfusion** in skin, tendons, ligaments, and other soft tissues.
- **Reduced gastrointestinal** and **systemic inflammation,** with **support for epithelial repair, mucosal integrity,** and **digestive comfort**.
- **Immunomodulatory support** potentially **promoting a more balanced immune profile** via antiviral, Th1, NK-cell, T-cell regulatory, and inflammation-resolving pathways.

- **Cognitive and neurological support**, including potential improvements in focus, neuroplasticity, and resilience to stress.
- **Improved cellular energy** and **recovery** through mitochondrial support, $NAD^+$ restoration, autophagy, and mitophagy.
- **Improved exercise performance** and buffering capacity during high-intensity activity.
- **Selective clearance of senescent cells**, potentially **reducing SASP-associated inflammation** and other signals that impair tissue repair.
- **Potential longevity and circadian support**, including proposed telomeric and pineal effects (relies substantially on preclinical and weak, regional clinical research).
- **Potential support for prostate-tissue repair**, local inflammation control, and microcirculation; supporting evidence remains limited.

## Metabolic Benefits

The stack is designed to produce **robust metabolic improvement** by combining **appetite suppression**, **incretin-mediated insulin support**, and a plausible **increase in energy expenditure**. Retatrutide provides strong GLP-1/GIP-driven reductions in appetite and postprandial glucose. Its glucagon-receptor activity may plausibly contribute to increased energy expenditure and fat oxidation. Retatrutide substantially reduces body weight and liver fat and **can normalize liver fat** in many people with metabolic dysfunction-associated steatotic liver disease. Metformin **improves hepatic insulin sensitivity** and **reduces gluconeogenesis**. Creatine and beta-alanine **support high-intensity exercise capacity**, while NMN and ALCAR plausibly support **cellular energy metabolism**. Together, these effects plausibly support metabolic flexibility. PQQ may **support mitochondrial-biogenesis signaling** and **redox regulation**, plausibly contributing to cellular energy function and metabolic flexibility. Together, these agents plausibly provide additive reductions in caloric intake, improvements in glycemic control, and increased energy expenditure, favoring body-composition improvement.

## GH Benefits

The GH-axis regimen (Ipamorelin + CJC-1295 no-DAC before sleep and Tesamorelin + Ipamorelin upon waking, both 12 weeks on and 6 weeks off) should create a **measurable increase in pulsatile GH secretion** and a **rise in circulating IGF-1** over baseline. (CJC-1295 no-DAC refers to modified GRF 1-29.) In practical terms, this is likely to **shift average GH/IGF-1 markers toward values more typical of middle-aged adults rather than untreated older adults**.

## Regenerative Benefits

This stack combines peptides whose proposed **mechanisms act at complementary points in the repair cascade** and could plausibly **provide additive signaling for matrix remodeling, inflammation control, and tissue repair**. **GHK-Cu upregulates collagen and proteoglycan synthesis** and modulates MMP/TIMP balance to rebuild extracellular matrix; **TB-500 enhances actin dynamics, cell migration, and angiogenesis** to mobilize reparative cells and revascularize injured tissue; **BPC-157 stimulates local growth-factor signaling and epithelial restitution** to stabilize microcirculation and accelerate tendon, ligament, and gut mucosal healing; and **KPV reduces NF-κB-driven cytokine release and supports epithelial barrier repair**, lowering the inflammatory milieu that impairs regeneration.

**ARA-290 likely supports microvascular preservation** and repair through tissue-protective and anti-inflammatory signaling.

The GH-axis agents (Ipamorelin + CJC-1295 no-DAC and Tesamorelin) provide systemic support through increased pulsatile GH/IGF-1 signaling, promoting p**rotein synthesis, satellite-cell activation,** and **anabolic repair processe**s.

*Prostamax (KEDP) is speculative. Some practitioners use it with the aim of supporting local prostate tissue regulation and microcirculation, reducing inflammation, and promoting tissue homeostasis.*

## Immune Benefits

This stack enhances an ageing immune system by combining **Tα1-driven activation of dendritic cells, NK cells, and interferon-regulated Th1 pathways.** Tα1's long-tail signaling **maintains elevated antiviral gene expression**, **improved antigen presentation**, and **stronger cytotoxic readiness**.

**Thymulin has probable effects on thymocyte maturation**, **T-cell activation thresholds**, and **proinflammatory cytokines** such as TNF-α, IL-1β, and IL-6. Its relatively short signaling window could plausibly allow repeated modulation of T-cell sensitivity, thymic-hormone signaling, and cytokine balance, contributing to a more coordinated, low-noise immune environment.

Because Thymulin is **proposed to support thymic and T-cell regulatory balance**, while Thymosin α1 **modulates dendritic-cell, NK-cell, and interferon-associated Th1 signaling**, their distinct but potentially overlapping mechanisms could plausibly provide complementary immunomodulatory signaling. The intended result is **better coordination**

**between immune activation and inflammatory regulation**, **supporting antiviral and cytotoxic readiness while limiting excessive inflammatory noise**.

## Sleep Benefits

**Epitalon, DSIP,** and **L-Glycine,** combined with **CJC-1295 no-DAC + Ipamorelin**, target distinct but potentially overlapping aspects of sleep regulation and regeneration. Together, these agents could plausibly support a sleep environment by **involving circadian timing, sleep-related neuroendocrine regulation, thermoregulatory cooling, nocturnal GH signaling,** and **overnight recovery**.

**Tadalafil** improves BPH-associated lower-urinary-tract symptoms and may **reduce nocturia-related sleep disruption**. Beta-sitosterol provides plausible complementary symptom support, while Prostamax remains speculative.

## Healthspan Benefits

A ***conserved pathway of ageing*** is a biological process that shows up across many species, changes predictably with age, and when experimentally modified can speed up or slow down ageing. These pathways represent the deep, evolutionarily preserved mechanisms that drive cellular decline, tissue dysfunction, and loss of resilience over time.

If a therapy targets a **conserved pathway of ageing** and ***if interventions on that pathway reliably slow ageing in model organisms***, then modifying that same pathway in humans is **plausibly healthspan-promoting**, even if direct human longevity data are not yet available.

The following eight conserved pathways are the **dominant, consensus model** used across ageing biology, geroscience, biotech, and translational longevity research.

**Mitochondrial dysfunction** — age-related decline in mitochondrial energy production, increased ROS, and impaired metabolic flexibility.

**Deregulated nutrient sensing (mTOR/AMPK/insulin/IGF)** — disrupted balance between growth and repair signaling that skews metabolism toward ageing rather than maintenance.

**Cellular senescence and SASP** — accumulation of damaged, non-dividing cells that secrete inflammatory factors and impair tissue function.

**Genomic instability and DNA repair** — increasing DNA damage and reduced repair capacity that undermine cellular integrity and accelerate ageing.

**Epigenetic drift** — progressive, disorganized changes in DNA methylation and chromatin structure that alter gene expression and degrade cellular identity.

**Stem cell exhaustion and regenerative niche** — depletion or dysfunction of tissue-specific stem cells, reducing the body's ability to repair and regenerate.

**Microvascular decline and endothelial dysfunction** — impaired blood flow, nitric-oxide signaling, and capillary density that limit oxygen and nutrient delivery.

**Dysbiosis and barrier dysfunction** — age-related shifts in the gut microbiome and weakening of epithelial barriers that increase inflammation and metabolic stress.

The following table evaluates how comprehensively this stack mechanistically targets these conserved pathways of ageing. Coverage ratings reflect the number, relevance, and complementary mechanisms of the interventions directed at each pathway.

| **Conserved pathway** | **What you are doing to address it** | **Coverage** |
|---|---|---|
| **Chronic inflammation (inflammaging)** | KPV; ARA-290; omega-3 (DHA/DPA); metformin; Thymosin-α1; Thymulin | **Excellent** |
| **Proteostasis and autophagy** | 16-hour TRF; spermidine; apigenin; metformin; NMN; resistance training | **Excellent** |
| **Mitochondrial dysfunction** | NMN; ALCAR; CoQ10; creatine HCl; Beta-alanine; PQQ; Retatrutide; exercise | **Excellent** |
| **Deregulated nutrient sensing (mTOR/AMPK/insulin/IGF)** | Metformin; Retatrutide; TRF; timed GH-axis cycling | **Good** |
| **Cellular senescence and SASP** | Thymosin α1; Thymulin; apigenin; immune surveillance; FOXO4-DRI senolytic pulses (Q48H ×3 every 2 years) | **Moderate** |
| **Genomic instability and DNA repair** | NMN (NAD+ support); good sleep/circadian hygiene; annual Epitalon + Crystagen + Pinealon window | **Good** |

| | | |
|---|---|---|
| **Epigenetic drift** | Annual Epitalon; resistance training 2×/week; sleep/circadian practices; NMN | **Moderate** |
| **Stem cell exhaustion and regenerative niche** | GH-axis peptides (Ipamorelin/CJC-1295 no-DAC, Tesamorelin); GHK-Cu; TB-500; BPC-157; exercise | **Moderate** |
| **Microvascular decline and endothelial dysfunction** | ARA-290; tadalafil 5 mg daily; omega-3; Retatrutide metabolic benefits | **Excellent** |
| **Dysbiosis and barrier dysfunction** | BPC-157; KPV; metformin (indirect); whole-food vegan diet | **Moderate** |

## Stack Design - Synergies

- **PM Ipamorelin + CJC-1295 no-DAC; AM Tesamorelin + Ipamorelin:** The evening **Ipamorelin/CJC-1295 no-DAC pairing is timed to coincide with the body's nocturnal GH rhythm**, with the intended aim of supporting sleep-associated recovery signaling. The **morning Tesamorelin + Ipamorelin combination is intended to provide a separate GH pulse during the daytime fasting period, when its lipolytic signaling may complement metabolic goals**. Tesamorelin (GHRH-R agonist) and Ipamorelin (GHSR-1a agonist) act through different receptors, providing a mechanistic rationale for a complementary secretory response. This timing separation plausibly aligns recovery-oriented nighttime signaling with metabolically oriented daytime signaling.
- **Retatrutide + Tesamorelin:** Both agents reduce liver fat, and tesamorelin reduces visceral abdominal fat. Retatrutide's glucagon-receptor activity plausibly increases energy expenditure and fat oxidation. Together, the agents plausibly **provide additive effects on hepatic and visceral fat reduction**.
- **GHK-Cu + TB-500 + BPC-157 + KPV + ARA-290:** These regenerative peptides target complementary pathways involving **collagen synthesis, actin remodeling, angiogenesis, epithelial repair, microvascular protection, tissue-protective signaling, and anti-inflammatory signaling**, which could plausibly provide additive repair signaling.
- **Epitalon, DSIP, and L-Glycine:** This combination could plausibly support sleep through **complementary and potentially overlapping effects on circadian signaling, sleep-related neuroendocrine pathways, thermoregulatory cooling, and glycine-mediated neurotransmission**.

- **NMN + ALCAR + CoQ10 + Creatine + Beta-alanine + Apigenin + PQQ:** These mitochondrial and energy supplements have plausibly complementary mechanisms that **improve cellular energy capacity** by supporting NAD+ pools, electron-transport-chain function, antioxidant defenses, fatty-acid metabolism, rapid ATP regeneration, PQQ-driven mitochondrial-biogenesis signaling, and intracellular buffering.
- **Semax + NMN/ALCAR**: Semax's probable neurotrophic effects are plausibly complementary to the metabolic support provided by NMN and ALCAR, with the intended goals of **supporting cognitive performance** and **neuronal metabolic resilience**.
- **Thymosin α1** contributes immune-activation signaling through dendritic-cell priming, NK-cell cytotoxicity, and interferon-associated Th1 pathways, while **Thymulin** contributes thymic and T-cell regulatory signaling. Because these effects are distinct but potentially overlapping, the **combination could plausibly improve coordination between immune activation and inflammatory regulation**.
- **Tadalafil + Beta-sitosterol + Prostamax**: Daily tadalafil improves lower-urinary-tract symptoms associated with BPH. Beta-sitosterol provides plausible complementary symptom support, while Prostamax remains a speculative organ-specific intervention. Together, they form a **plausible prostate-support strategy** involving symptom control, cGMP-mediated smooth-muscle effects, and possible local microcirculatory support.
- **ARA-290 + tadalafil + omega-3:** These agents have mechanistically complementary effects that could plausibly support microvascular and endothelial function.
- **FOXO4-DRI, Spermidine, intermittent fasting, NMN, apigenin,** and **resistance training** may plausibly support autophagy and proteostasis through complementary nutritional, metabolic, and exercise-related mechanisms.
- **Thymosin α1** and **Thymulin** plausibly support immune surveillance around the FOXO4-DRI senolytic pulses.

## Stack Design - Compatibility

These stack items are in different primary physiologic lanes and their differing primary mechanisms are plausibly compatible, while their downstream effects may overlap and produce complementary, additive effects.

**The mitochondrial lane overlap:** The mitochondrial and energy supplements are **mechanistically compatible**: **ALCAR** supports mitochondrial substrate entry and acetyl-CoA availability, **CoQ10** enhances ETC efficiency, **creatine** buffers cellular ATP and stabilizes energy supply, and **Beta-alanine** raises muscle carnosine to improve high-intensity performance; **apigenin** is a polyphenol that modulates signaling pathways

(AMPK/SIRT, antioxidant responses); **omega-3s** improve membrane composition, reduce inflammation, and promote mitochondrial resilience. None of these directly block each other's core actions.

**The GH lane overlap**: **Tesamorelin**, **Ipamorelin**, and **CJC-1295 no-DAC** all modulate GH/IGF-1 and require coordinated dosing. Separating **morning** Tesamorelin + Ipamorelin from **evening** Ipamorelin + CJC-1295 no-DAC plausibly reduces overlap and aligns their stimulation with daytime fasting metabolism and nocturnal recovery.

**The immune lane overlap**: **Thymulin** and **Thymosin α1** act through distinct aspects of thymic and immune regulation. Their combination is intended to provide complementary immunomodulatory signaling and should be coordinated as a shared immune-modulation lane.

**The sleep lane overlap**: **Epitalon, DSIP, and L-Glycine** act through distinct but potentially overlapping mechanisms. Their combination is intended to coordinate pineal and circadian signaling, sleep-related neuroendocrine pathways, and glycine-mediated thermoregulatory and neurotransmitter effects.

The annual **Epitalon + Crystagen + Pinealon longevity window** is intentionally run with other peptides paused (except GLP-1s) to create a clear biologic window and reduce confounding interactions.

Separating **FOXO4-DRI senolytic pulses** from **GH-axis on-cycles** and repair peptides plausibly reduces overlap between senolytic and anabolic or regenerative signaling.

## Stack Design - Conservative Dosing and Cycles

This stack uses a **conservative, low dose of retatrutide**, divided across the week, with the intent of retaining metabolic benefit without encouraging weight loss.

**GH secretagogues** are cycled 12 weeks ON and 6 weeks OFF, a commonly used empirical schedule that plausibly limits continuous GH-axis stimulation and helps preserve responsiveness.

By **alternating GLOW / KLOW and GHK-Cu every 5 weeks**, we reduce the uncertainties of continuous TB-500**.**

Overall, many items are **intentionally dosed well below community-reported maxima to prioritize safety** and tolerability.

# Cautions

Informed consent takes on a serious meaning and role in the context of gray-market sourcing, peptides that are not FDA approved, prescriptions that are used in off-label contexts, and protocols that call for near-continuous use.

# Specific Cautions

- Active or recent malignancy or elevated cancer risk: GH-axis stimulants (Ipamorelin, CJC-1295 no-DAC, Tesamorelin) and regenerative or potentially angiogenic peptides (GHK-Cu, TB-500, BPC-157) warrant particular caution and oncology review before use.
- Marked baseline IGF-1 elevation or pituitary disease: GH-axis stimulants warrant caution and should not be initiated until IGF-1 testing and appropriate endocrine evaluation are complete.
- Uncontrolled diabetes or insulin deficiency: GLP-1/GIP/glucagon agents and GH-axis stimulants can alter glycemia; unstable diabetes warrants particular caution, and type 1 diabetes requires specialist oversight.
- Significant hepatic impairment: Metabolic and hepatic-targeting agents warrant particular caution; decompensated liver disease requires specialist oversight.
- Severe renal impairment: Clearance or tolerability of some components may be altered; advanced chronic kidney disease warrants individualized review, including component-specific dosing and monitoring.
- Pregnancy and breastfeeding: Peptides and metabolic agents warrant particular caution and generally should be avoided during pregnancy and lactation unless specifically recommended by an appropriate specialist.
- Active autoimmune disease or immunosuppressive treatment: Immune-modulating peptides warrant caution because they may alter immune activity; coordinate their use with the treating specialist.
- History of pancreatitis or severe gastroparesis: GLP-1/GIP agents warrant particular caution.
- Hypotension and syncope risk: Low-dose daily tadalafil and other vasodilatory agents can lower blood pressure; use caution with antihypertensives or alpha-blockers. Nitrates should not be combined with tadalafil. Review potent CYP3A4 inhibitors, which can raise tadalafil exposure.

# Monitoring

**Baseline (before starting or continuing):** fasting IGF-1 (use same lab, same assay, same time of day each time), fasting glucose, HbA1c, fasting insulin (HOMA-IR), CMP (AST, ALT, bilirubin, creatinine, electrolytes with magnesium), CBC with differential, lipid panel, TSH + free T4, serum copper + ceruloplasmin + zinc (for GHK-Cu), PSA and age-appropriate cancer screening.

**At 3 months, then every 6 months while on therapy:** IGF-1, HbA1c, fasting glucose, CMP, lipid panel, CBC; copper, ceruloplasmin, zinc; thyroid panel, PSA. **Off-cycle check:** IGF-1 at 4–6 weeks into washout to confirm return toward baseline.

**Action Thresholds**

- Persistent IGF-1 above +3 standard deviation scores (SDS) for age: pause GH-axis stimulants and evaluate.
- Fasting blood glucose increase greater than 20 mg/dL: reassess metabolic agents.
- A1c increase of 0.3 percentage points or more: reassess metabolic agents.
- Aspartate aminotransferase (AST) or alanine aminotransferase (ALT), markers of liver injury, greater than twice the laboratory's upper limit of normal (ULN): repeat testing, investigate other causes, and review retatrutide, tesamorelin, tadalafil, metformin, supplements, and experimental peptides. Consider pausing the component most closely associated in time with the abnormality.
- Serum creatinine increase greater than 30% from baseline: repeat testing, assess hydration and estimated glomerular filtration rate (eGFR), and consider whether creatine supplementation is raising serum creatinine without reducing kidney function. Review metformin and tadalafil because kidney function affects their use, and review retatrutide if vomiting, diarrhea, or dehydration is present.

# Biological Mechanisms

Retatrutide — a triple agonist at GLP-1, GIP, and glucagon receptors. By adding glucagon-receptor activation to incretin signaling, it may plausibly increase systemic energy expenditure and fat oxidation through cAMP-mediated lipolysis and mitochondrial substrate switching, while the GLP-1/GIP components support appetite suppression and postprandial insulin dynamics. The combined activity produces substantial reductions in body weight and liver fat.

Metformin — a biguanide insulin sensitizer. It lowers hepatic glucose output through several complementary mechanisms, including effects on mitochondrial complex I and cellular energy state, AMPK-dependent and AMPK-independent signaling, hepatic redox balance, glucagon signaling, and gut-mediated pathways. These actions reduce gluconeogenesis and support improved glycemic control, while secondary effects on lipid metabolism and inflammation contribute to its broader metabolic benefits.

Ipamorelin — a selective ghrelin receptor (GHSR) agonist; CJC-1295 no-DAC — a GHRH receptor potentiator. Ipamorelin mimics ghrelin to trigger pulsatile GH release via GHSR-mediated signaling, while CJC-1295 no-DAC supports GHRH-driven pituitary stimulation; together they are intended to augment nocturnal GH signaling, with potential downstream effects on IGF-1, protein synthesis, and tissue-repair pathways. Cycling is a commonly used empirical strategy that plausibly limits continuous receptor exposure and helps preserve responsiveness.

Tesamorelin — a GHRH analogue that binds the pituitary GHRH receptor and increases endogenous pulsatile GH secretion. It reduces visceral abdominal fat and liver fat. These effects are plausibly mediated through GH-dependent lipolysis and hepatic lipid metabolism.

GHK-Cu — a copper-binding tripeptide (glycyl-L-histidyl-L-lysine–copper). It likely supports matrix remodeling, wound healing, and dermal and soft-tissue repair through plausible effects on collagen and proteoglycan synthesis, MMP/TIMP balance, inflammatory signaling, and oxidative-stress regulation.

TB-500 — a synthetic peptide corresponding to an actin-binding region of thymosin β4. It plausibly supports tissue repair through effects on actin-cytoskeleton remodeling, cell migration, angiogenic signaling, and extracellular-matrix reorganization.

BPC-157 — a synthetic gastric pentadecapeptide. It likely supports tendon, ligament, muscle, and gastrointestinal tissue healing. These effects are plausibly mediated through VEGFR2 and nitric-oxide signaling, angiogenesis, inflammatory modulation, epithelial repair, and stabilization of local microcirculation.

KPV (Lys-Pro-Val) — an anti-inflammatory tripeptide corresponding to the C-terminal sequence of α-melanocyte-stimulating hormone. It likely reduces intestinal inflammation and supports mucosal-barrier recovery through plausible effects involving PepT1-mediated cellular uptake, inhibition of NF-κB and MAPK signaling, and reduced pro-inflammatory cytokine production.

Omega-3 (DHA and DPA) — long-chain marine n-3 fatty acids that integrate into cellular and neuronal membranes and support membrane structure and function. DHA reduces triglyceride levels. DPA contributes to bioactive lipid-mediator pathways and plausibly supports inflammation resolution and vascular function. Together, DHA and DPA plausibly support neuronal, endothelial, and metabolic health.

Thymosin α1 — a thymic peptide immunomodulator. It plausibly supports T-cell maturation, antigen-presenting-cell function, and immune surveillance by modulating innate and adaptive immune signaling, including Th1 and regulatory pathways, rather than acting as a broad immunosuppressant.

Epitalon — a tetrapeptide acting on pineal and circadian pathways; Crystagen — a thymic peptide. Epitalon modulates melatonin and circadian signaling and plausibly influences telomere-related pathways. Crystagen likely supports thymic signaling and T-cell homeostasis. Together, an annual course plausibly influences circadian biology and aspects of immune ageing. Longevity effects remain speculative.

Tadalafil — a selective PDE5 inhibitor that raises intracellular cGMP by preventing its breakdown, thereby potentiating nitric-oxide–mediated vasodilation; this improves endothelial function, microvascular perfusion, and tissue oxygenation. Chronic PDE5 inhibition also favorably modulates vascular inflammation and arterial stiffness and plausibly influences mitochondrial signaling and perfusion-dependent repair processes in prostate and erectile tissues, making daily low-dose tadalafil a plausible adjunct for healthspan-oriented microcirculatory and metabolic support.

Prostamax (KEDP) — a synthetic tetrapeptide (Lys-Glu-Asp-Pro) from the Khavinson 'bioregulator' family. Some practitioners use it as a prostate-targeted bioregulator with the aim of supporting prostate-tissue regulation, reducing local inflammation, improving microcirculation, and promoting repair. Its proposed chromatin and gene-expression mechanisms and clinical benefits remain speculative.

Semax — a synthetic peptide derived from ACTH fragment 4–10 and delivered intranasally. It modulates neurotrophic and monoaminergic signaling, with proposed effects on BDNF-related pathways and synaptic plasticity. A substantial regional human evidence base and longstanding Russian clinical use support neuroprotective effects in cerebrovascular and neurologic settings, with reported improvements in attention and operative memory. Semax is commonly administered in short courses, and cycling is widely reported in clinical and practitioner use as necessary to maintain responsiveness.

L-glycine — a nonessential amino acid that supports collagen synthesis, serves as an inhibitory neurotransmitter in the spinal cord and brainstem, and acts as a key glutathione precursor and one-carbon metabolism substrate, thereby contributing to antioxidant defenses and cellular repair. Mechanistically, oral glycine modulates central NMDA/glycine receptor signaling and promotes peripheral vasodilation that lowers core body temperature, a thermoregulatory change that facilitates sleep onset. Oral glycine at the commonly studied dose improves sleep quality, reduces sleep latency, and improves next-day alertness.

Beta-sitosterol — a plant sterol that reduces intestinal cholesterol absorption and modestly lowers LDL cholesterol. It plausibly improves urinary symptoms of benign prostatic hyperplasia through anti-inflammatory and membrane-modulating effects in prostate tissue.

β-Nicotinamide mononucleotide (NMN) — an NAD+ precursor that is converted intracellularly to NAD+. Oral NMN increases NAD-related metabolites. This plausibly supports redox balance, sirtuin-dependent signaling, mitochondrial function, and DNA-repair pathways involved in cellular energy metabolism and resilience.

Acetyl-L-Carnitine (ALCAR) — an acetylated carnitine derivative and acetyl-group donor. ALCAR improves pain and nerve function in diabetic peripheral neuropathy. By contributing to the cellular carnitine pool and acetyl-CoA buffering, it plausibly supports fatty-acid utilization, neuronal energetics, and skeletal-muscle mitochondrial function.

CoQ10 — a mitochondrial electron-transport cofactor that acts as a mobile electron carrier within the inner mitochondrial membrane, accepting electrons from Complexes I and II and transferring them to Complex III. This role supports ATP production. Its redox activity plausibly supports antioxidant defenses, mitochondrial membrane stability, and cardiovascular and metabolic function.

Creatine HCl — a creatine salt that plausibly supports cellular energy buffering by supplying creatine for phosphocreatine production and rapid ATP regeneration.

Beta-alanine — a carnosine precursor. By increasing intramuscular carnosine concentrations, beta-alanine improves intracellular pH buffering during high-intensity exercise, reduces fatigue, and enhances performance and training capacity. Combined with creatine and mitochondrial support, it plausibly provides complementary support across muscular energy systems.

Apigenin — a naturally occurring flavone with plausible antioxidant and anti-inflammatory activity. It plausibly influences cellular-stress and senescence-related signaling through pathways involving NF-κB, AMPK, SIRT1, and NAD+ metabolism, with downstream support for mitochondrial, endothelial, and neuronal resilience.

PQQ — a redox-active quinone that plausibly supports mitochondrial-biogenesis signaling. It plausibly influences CREB and PGC-1α signaling, mitochondrial number, electron-transport function, and oxidative-stress responses. These mechanisms plausibly support neuronal resilience and cellular energy function and complement NMN, ALCAR, and CoQ10 by supporting mitochondrial capacity.

Minoxidil — a hair-growth treatment that increases hair density and hair weight in androgenetic alopecia. Its effects are plausibly mediated through ATP-sensitive potassium-channel activity, earlier entry of resting follicles into anagen, prolongation of the growth phase, and modulation of VEGF, prostaglandin E2, and Wnt/β-catenin signaling.